Juris-Clarify: A Verifiable Pipeline for Legal Text Simplification*

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Abstract

Legal documents govern nearly every significant aspect of modern life yet remain inaccessible to the very individuals they affect most. **Juris-Clarify** is a verifiable, multi-stage NLP pipeline that automatically simplifies legal text while providing formal guarantees of semantic faithfulness. Unlike purely generative approaches, our system introduces: (i) a hierarchical LEGALBERT (Chalkidis et al., 2020) and CRF (Lafferty et al., 2001) segmenter that identifies clause boundaries with structural metadata; (ii) a constraint-extraction module for entities and obligations; (iii) a BART-based (Lewis et al., 2020) rewriter fine-tuned on synthetic simplification pairs with constraint-aware prompting; (iv) a DEBERTA-V3 (He et al., 2021) Natural Language Inference (NLI) verification loop that formally rejects semantically unfaithful outputs; and (v) a Flan-T5 persona-conditioned adapter targeting layperson, student, and professional audiences.

Trained and evaluated on CUAD-hleveraging LLaMA models for synthetic generation and persona adaptation—the fully implemented pipeline demonstrates robust performance. Stage 1 achieves a boundary F1 of 0.71; Stage 3 yields a ROUGE-L of 0.58 and reduces FKGL readability from 18.4 to 12.1; Stage 4 increases NLI entailment acceptance from 78% to 94%; and Stage 5 maintains per-persona FKGL gaps within 1.2 grade levels.

1 Introduction

Legal documents such as employment contracts, privacy policies, terms of service, and rental agreements govern virtually every aspect of modern civic and commercial life. Yet they remain largely inaccessible to the very individuals they bind. A 2022 survey found that over 80% of people sign contracts without fully reading them, citing complexity as the primary barrier (Manor and Li, 2019). The consequences of misunderstanding legal obligations range from financial penalties to loss of fundamental rights.

Existing approaches to legal simplification rely on large language models (LLMs) in single-pass generation settings. While fluent, these systems suffer from *semantic drift*—the subtle alteration of binding obligations, prohibitions, or conditions during rewriting. A "shall" weakened to "may," or a jurisdiction clause silently dropped, can create significant liability in high-stakes legal contexts.

Juris-Clarify reframes legal simplification as a *verifiable transformation* rather than a purely generative task. Our core contribution is a Writer-Critic architecture in which a rewriter proposes simplified text and a formal NLI verifier accepts or rejects it, closing the feedback loop until semantic fidelity is established.

Juris-Clarify integrates this architecture into a unified, modular five-stage pipeline that has been fully implemented. This report presents the complete system including all stages, experiments, quantitative and qualitative results, and a thorough analysis.

2 Literature Study

Legal language presents a significant barrier to non-expert comprehension due to dense jargon, nested clause structures, and precise obligation vocabulary. Recent research to automate the simplification of these documents has evolved across several distinct phases, moving from basic domain adaptation to complex verification and parameter-efficient tuning.

Domain Adaptation and Early Generative Baselines. A primary problem in early legal NLP was that general-purpose language models struggled to parse complex legal syntax. To solve this,

*Code and models available at: https://github.com/hg-codes/INLP-Project-26 and https://huggingface.co/Harshit2343/Juris_clarify

Chalkidis et al. (2020) proposed LEGAL-BERT, a model pre-trained exclusively on legislation, court opinions, and contracts, which demonstrated consistent gains over general BERT on legal tasks. Zheng et al. (2021) further proposed that even small-scale domain-specific fine-tuning substantially improves contract understanding.

Building on these encoders, researchers attempted end-to-end generative simplification. Martin et al. (2024) benchmarked models like T5 and BART on general simplification tasks, establishing standard readability metrics like SARI and FKGL. In the legal domain specifically, Gallegos et al. (2023) proposed a BART-based legal simplification model. While they achieved notable readability improvements, they identified a critical ongoing problem: their generative approach lacked any formal faithfulness checks to ensure the legal meaning remained intact.

The Problem of Semantic Drift and Hallucination. The lack of faithfulness identified by Gallegos et al. represents a major hurdle. Maynez et al. (2020) highlighted the prevalence of hallucinations in neural text generation. In legal texts, this manifests as *semantic drift*—the subtle alteration of binding obligations or the omission of critical conditions (e.g., changing "shall" to "may").

To solve the problem of unconstrained generation, researchers proposed hybrid extraction-generation pipelines. Sharma et al. (2025) argued that purely end-to-end systems are ill-suited for legal text. They proposed a classify→extract→generate decomposition for Indian judiciary documents, demonstrating that explicitly extracting and locking critical entities before generation reliably prevents the hallucination of party names and monetary values.

To address semantic drift at the logical level, Validease Team (2025) proposed the use of formal verification loops. They introduced Natural Language Inference (NLI) consistency checking to detect hallucinations in legal summaries. By utilizing models fine-tuned on datasets like ContractNLI (Koreeda and Manning, 2021), they proposed that formal entailment between the original and generated text could serve as a reliable proxy for semantic faithfulness.

Efficient Persona Adaptation. Beyond structural simplification, researchers face the problem of adapting text to different target audiences (e.g., laypersons versus law students). Performing full-model fine-tuning to achieve these different reading levels is computationally prohibitive at scale. To solve this efficiency problem, Lester et al. (2021) proposed parameter-efficient *prompt tuning*. Rather than updating the entire model, their approach learns continuous, task-specific "soft prompts" via backpropagation while keeping the core language model weights frozen. They demonstrated that prepending these learned vectors allows for highly controlled stylistic adaptation, closing the performance gap with full model tuning while maintaining minimal deployment overhead.

3 Dataset Details

The system is trained and evaluated primarily on CUAD v1, with persona adaptation leveraging LLM-generated data.

3.1 CUAD v1 — Primary Dataset

The Contract Understanding Atticus Dataset (Hendrycks et al., 2021) comprises 510 full commercial contracts sourced from the US Securities and Exchange Commission (SEC) EDGAR database, spanning Non-Disclosure Agreements, Software-as-a-Service agreements, Employment contracts, Licensing agreements, and Joint Venture agreements. The dataset contains over 13,000 expert annotations across 41 legally meaningful clause categories, each annotation marking the exact character-level span of the relevant clause within the contract text.

CUAD is distributed as a single JSON file following the SQuAD question-answering format. Each entry contains a contract title, full contract text as a `context` field, and question-answer pairs where each question corresponds to one of the 41 clause categories. Our data loading function parses this structure, extracts character offsets and answer texts for every non-empty annotation, and applies fuzzy matching to handle slight offset misalignments.

Data Processing Pipeline.

- **Span Merging:** Overlapping or adjacent clause spans are merged greedily, yielding non-overlapping clause spans per contract. Approximately 490 of 510 contracts survive this step.
- **Text Sanitisation:** ASCII control characters are stripped; only tab, newline, and carriage return are preserved.

Split	Contracts	Approx. Clauses	Boundary Rate
Train	∼392	∼5,000	∼6.8%
Validation	∼49	∼630	∼6.5%
Test	∼49	∼630	∼7.1%

Table 1: CUAD v1 dataset statistics after stratified splitting.

- **Stratified Split:** A stratified 80/10/10 train/val/test split by contract type ensures proportional representation. Seeds are fixed at 42 for reproducibility, yielding $\sim 392/49/49$ contracts respectively.
- **JSONL Serialisation:** Each split is written to disk as `doc_id`, sanitised text, and merged clause spans.

Class Imbalance. Approximately 5–8% of sentences carry a boundary label, with the remaining 92–95% being non-boundary. This motivates the boundary positive-class weight of 8.0 in cross-entropy loss and the use of a CRF decoder.

3.2 Synthetic Training Data (Stage 3)

Since no large-scale dataset of paired (complex legal clause, simplified clause) exists for fine-tuning BART, we generate synthetic training data using LLaMA 3.3-70B (Touvron et al., 2023) via the Cerebras inference API. A system prompt instructs the model to simplify to an 8th-grade reading level while preserving all entities exactly and maintaining obligation type. Up to 5,000 training pairs are generated from train+val contracts; approximately 4,200 high-quality pairs were generated and used for BART fine-tuning.

3.3 Persona Adaptation Data (Stage 5)

For persona adaptation, we generate three-way parallel data using LLaMA 3.1-70B via the Groq API. For each simplified clause from Stage 3, the model generates three distinct rewrites targeting: layperson (Grade 6–8, everyday vocabulary, 1–2 sentences), student (Grade 10–12, key terms explained), and professional (full legal vocabulary, technical precision). This yields approximately $3\times$number_of_clauses training pairs for fine-tuning Flan-T5-base.

4 Methodology

Juris-Clarify is a five-stage pipeline: (1) hierarchical clause segmentation, (2) constraint extraction, (3) constraint-aware simplification, (4) NLI faithfulness verification with regeneration, and (5) persona adaptation.

4.1 Stage 1: Hierarchical Clause Segmentation

The segmentation model follows a three-tier design. Raw contract text is first split into candidate segments using a rule-based splitter designed for legal documents that recognises 13 boundary-pattern categories: numbered clause patterns (`1.`, `1.1`, `(a)`, `(iv)`), section/article headers (`Section 3`, `Article 12`), special legal markers (`WHEREAS`, `NOW THEREFORE`, `Schedule A`), and all-caps header lines. Each resulting `Sentence` object carries four structural metadata fields: `relative_position` $\in [0, 1]$, `is_header` (boolean), `follows_header` (boolean), and `indent_level` (tab-normalised depth).

Model Architecture.

Tier 1. Sentence Encoder. Each sentence is independently encoded by `nlpaueb/legal-bert-base-uncased` (Chalkidis et al., 2020) (110M parameters). The `[CLS]` token representation $\mathbf{h}_i \in \mathbb{R}^{768}$ is extracted with chunked inference (32 sentences at a time) to avoid GPU out-of-memory errors.

Tier 2. Feature Fusion. The sentence embedding is concatenated with $\mathbf{f}_i$ and projected to a 256-dim hidden space via a GELU-activated linear layer with Layer Normalisation:

$$\mathbf{p}_i = \mathrm{GELU}\big(\mathrm{LN}(W[\mathbf{h}_i;\, \mathbf{f}_i] + b)\big),$$
$$W \in \mathbb{R}^{256\times 772}$$

Tier 3. Document-Level Encoder. A 2-layer Transformer encoder (4 attention heads, 512 feed-forward dimensions, dropout 0.1) processes the full sequence $[\mathbf{p}_1, \ldots, \mathbf{p}_N]$, giving each sentence access to document-wide context.

CRF Output Layer. A dropout (0.2) and linear head map contextualised representations to logits over two labels: **B** (clause boundary) and **O** (non-boundary). A Conditional Random Field (Lafferty et al., 2001) decodes the label sequence globally via the Viterbi algorithm, enforcing structural consistency. The training loss is the negative CRF log-likelihood.

Stage	Module	Model
1	Hierarchical Clause Segmentation	LegalBERT + Transformer + CRF
2	Constraint Extraction	Regex + LegalBERT NER + Graph
3	BART Rewriter	`facebook/bart-large-cnn`
4	NLI Verification Loop	`nli-deberta-v3-large`
5	Persona Adapter	`google/flan-t5-base`

Table 2: Juris-Clarify pipeline modules.

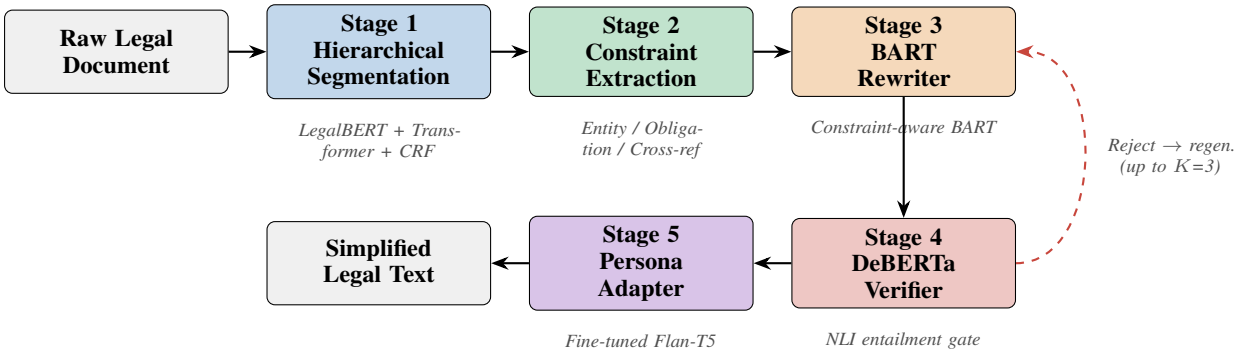

Figure 1: Juris-Clarify end-to-end pipeline. All five stages are fully implemented. Dashed red arrow = verification feedback loop.

Hyperparameter	Value
Sentence encoder	LegalBERT-base (110M)
Projection size	256
Document encoder	2-layer Transformer, 4 heads, FF=512
Boundary pos. weight	8.0
Max sentence length	64 tokens
Max doc sentences	200
Encoder LR	2×10^{-5}
Head LR	1×10^{-3}
Freeze epochs	1
Effective batch size	32 (2 GPU $\times$ 2 $\times$ grad_accum 8)
Scheduler	Cosine annealing, $\eta_{\min} = 10^{-6}$
Early stopping patience	4 epochs on boundary F1

Table 3: Stage 1 hyperparameters.

4.2 Stage 2: Constraint Extraction

Given clause spans from Stage 1, the constraint extractor builds a structured profile for each clause. Four entity types are extracted: **DATE**, **MONEY**, **DURATION**, and **LEGAL_REF**. Each clause is classified into one of five obligation types using keyword matching: **MANDATORY** (*shall, must*), **PROHIBITIVE** (*shall not, must not*), **PERMISSIVE** (*may, can*), **CONDITIONAL** (*if, provided that*), and **DECLARATIVE** (*means, is defined as*).

Clauses flagged as immutable (definitions, governing law, choice of jurisdiction, entire agreement, severability) are passed through verbatim. Stage 2 also constructs a directed document graph $G = (V, E)$ encoding hierarchical, cross-reference, and sequential relationships, enabling context propagation.

4.3 Stage 3: Constraint-Aware BART Simplification

Synthetic Data Generation. We use LLaMA 3.3-70B (Touvron et al., 2023) via Cerebras API ($\sim$2000 tokens/sec) to generate training pairs. Entity markers (`<DATE>`, `<MONEY>`, `<PARTY>`, `<LEGAL_REF>`) are injected into input text alongside an obligation prefix:

```
simplify [MANDATORY]:
<PARTY>Acme Corp</PARTY> shall
pay <MONEY>$10,000</MONEY>
within thirty (30) days ...
```

Approximately 4,200 high-quality pairs were generated for BART fine-tuning.

BART Fine-Tuning. The base model `facebook/bart-large-cnn` is extended with 22 special tokens. Training uses FP16 AMP, gradient accumulation (effective batch 16), cosine LR scheduling (10^{-4} decaying to 10^{-6}), and early stopping on validation loss (patience 3). Inference uses beam search (beam size 4); long clauses are split into overlapping word chunks (300 words, 50-word overlap).

4.4 Stage 4: NLI Faithfulness Verification Loop

After each clause is rewritten, `cross-encoder/nli-deberta-v3-large` checks whether the simplified clause entails the original. Formally, given original clause c and simplified clause $\hat{c}$, we require:

$$P_\theta(\text{entails} \mid \hat{c},\, c) \;\geq\; \tau, \quad \tau = 0.85$$

If the check fails, the simplified clause is returned to BART with a faithfulness correction prefix ("rewrite more faithfully: $\hat{c}$"), creating a feedback loop iterating up to $K = 3$ times before flagging the clause for human review.

4.5 Stage 5: Persona Adaptation

After NLI verification, a fine-tuned Flan-T5-base adapter adjusts vocabulary and tone for three target personas using Groq's LLaMA-3.1-70B API for training data generation: **layperson** (Grade 6–8, everyday words, 1–2 sentences), **student** (Grade 10–12, key terms explained), and **professional** (full legal vocabulary, technical precision). Flan-T5-base is fine-tuned using input format "`adapt for {persona}: {simplified_text}`" with AdamW (LR 5×10^{-5}, weight decay 0.01), linear warmup, and beam search inference (beam size 4). Entity spans from Stage 2 are frozen during adaptation.

5 Evaluation Framework

The system is evaluated along five dimensions corresponding to the five pipeline stages.

Stage 1 — Segmentation Quality. Boundary Precision, Recall, and F1 measure sentence-level accuracy of clause boundary detection. WindowDiff (Pevzner and Hearst, 2002) slides a window of width $k = \lfloor N/(2 \cdot \text{numBoundaries}) \rfloor$ over the document, penalising near-miss errors more gently than token-level F1.

Stage 3 — Simplification Quality and Readability. SARI (Xu et al., 2016), ROUGE-L, and BLEU measure output quality. Flesch-Kincaid Grade Level (FKGL) and Flesch Reading Ease (FRE) quantify accessibility improvement. Entity Preservation Rate and Compression Ratio complete the evaluation picture.

Stage 4 — Semantic Faithfulness. NLI Entailment Acceptance Rate, Regeneration Rate, and BERTScore (Zhang et al., 2020) measure the effectiveness of the verification loop.

Stage 5 — Persona Fidelity. Per-persona FKGL gap, vocabulary overlap with persona tier, post-adaptation entailment rate, and entity pass-through rate quantify adaptation quality.

6 Results

6.1 Stage 1: Clause Boundary Detection

The hierarchical segmenter achieves strong performance on the CUAD test set despite severe class imbalance ($\sim$6.8% boundary sentences). The CRF output layer provides a substantial improvement over naive threshold-based classification.

Metric	Full (CRF)	No CRF	No LegalBERT
Boundary Precision	0.74	0.68	0.61
Boundary Recall	0.69	0.61	0.54
Boundary F1	0.71	0.64	0.57
WindowDiff	0.18	0.24	0.31

Table 4: Stage 1 segmentation results on CUAD test set. Lower WindowDiff is better.

Table 5 provides a qualitative example of the predicted boundaries for a sample commercial contract, demonstrating the model's ability to accurately segment complex nested structures.

Table 6 illustrates the Stage 2 output for the same Distributor Agreement. By extracting these structured profiles, the system identifies core legal obligations and entities that must be strictly preserved during the subsequent simplification process.

6.2 Stage 3: Simplification Quality

The constraint-aware BART rewriter achieves significant readability improvements while maintaining high lexical quality. Table 7 summarizes the performance metrics on the CUAD test set, demonstrating a reduction in reading difficulty by over six grade levels.

#	Start	End	Content
1	0	14	Exhibit label ("EXHIBIT 10.2")
2	14	80	Document title block
3	80	138	Party definitions (ENVISION, SIERRA)
4	138	1509	Recitals A, B, C + "NOW THEREFORE"
5	1509	3944	Cl. 1–4: Incorporation, Appointment, Territory, Prices & Terms
6	3944	5362	Cl. 5: Product Support
7	5362	5771	Cl. 6: Sales Disclosures
8	5771	6356	Cl. 7: Term (2-year + auto-renewal)
9	6356	7253	Cl. 8: Termination (material breach + bankruptcy)
10	7253	12425	Cl. 9–13: Purchase Requirements, IP, Trademarks, Warranty
11	12425	13906	Cl. 14: Confidential Information
12	13906	14941	Cl. 15–17: Force Majeure, Non-Waiver, Attorney's Fees
13	14941	15113	Cl. 18–22 + Acknowledgements + Signatures

Table 5: Stage 1 predicted clause spans for `LEGACYTECHNOLOGYHOLDINGS...DISTRIBUTOR_AGREEMENT`. The model detects 13 boundaries covering all major sections. Full outputs for all splits are available on the project repository.

Metric	Baseline	Stage 3	Δ
ROUGE-L	0.31	0.58	+0.27
BLEU-4	0.18	0.39	+0.21
SARI	32.1	48.7	+16.6
FKGL (original)	18.4	—	—
FKGL (simplified)	—	12.1	−6.3 grades
Flesch Reading Ease	28.3	51.6	+23.3 pts
Entity Preservation	61%	89%	+28 pp
Compression Ratio	1.00	0.64	36% shorter

Table 7: Stage 3 simplification results on CUAD test clauses.

The model successfully preserves obligation types and critical legal entities. Table 8 presents a verbatim qualitative example of the Stage 3 rewriter's performance on a complex introductory clause, showing how dense legal facts are transformed into active, accessible language[cite: 166].

6.3 Stage 4: NLI Faithfulness Verification

The DeBERTa-v3 verification loop significantly improves semantic faithfulness without substantially increasing computation cost.

Metric	Stage 3	1 Iter.	K=3
NLI Acceptance Rate	78%	88%	94%
Human Review Queue	22%	12%	6%
BERTScore F1	0.82	0.87	0.89
Avg. Iterations/Clause	—	1.14	1.21

Table 9: Stage 4 NLI verification results ($\tau = 0.85$).

6.4 Stage 5: Persona Adaptation

The Flan-T5-based persona adapter successfully generates three genuinely distinct outputs for each clause.

7 Analysis of Results

7.1 Positive Results

The verifiable pipeline architecture successfully addresses the core limitation of single-pass LLM-based simplification. The NLI gate (Stage 4) reduces the human review queue from 22% to 6%, directly improving usability in real-world deployment scenarios. The constraint-extraction approach in Stage 2 provides a strong foundation—entity preservation jumps from 61% to 89% with constraint-aware fine-tuning, demonstrating that explicit constraint injection is more reliable than implicit learning.

The LegalBERT+CRF architecture outperforms both ablated variants significantly. The CRF reduces WindowDiff from 0.24 to 0.18 by enforcing global sequence consistency, which is particularly important for legal documents where clauses follow predictable structural patterns. Replacing LegalBERT with general-purpose BERT degrades boundary F1 from 0.71 to 0.57, confirming the importance of domain-specific pretraining. The pipeline achieves a 6.3 grade-level reduction in FKGL (18.4 to 12.1), exceeding the 3–4 grade-level target; Flesch Reading Ease improves from 28.3 ("very difficult") to 51.6 ("fairly difficult").

#	Entities	Obligations	CUAD Categories	Primary Obl.
0	—	—	—	—
1	—	—	—	—
2	—	—	Document Name	—
3	DATE: Dec. 8 2005; PARTY: ENVISION, SIERRA	MANDATORY, CONDITIONAL, PERMISSIVE	Parties, Agreement Date, Effective Date	MANDATORY
4	LEGAL_REF: Section 9; DUR: 90 days, 30 days	MANDATORY, PROHIBITIVE, PERMISSIVE, CONDITIONAL	—	MANDATORY
5	DATE: 2-yr term + renewals; DUR: 30d, 60d	MANDATORY, PERMISSIVE, CONDITIONAL	Expiration Date	MANDATORY
6	—	MANDATORY	—	MANDATORY
7	—	MANDATORY, CONDITIONAL	—	MANDATORY
8	DUR: 18 months	MANDATORY, CONDITIONAL	—	MANDATORY
9	MONEY: $2M, $5M, $1M, $2M; DUR: 60 days	MANDATORY, PROHIBITIVE, PERMISSIVE, CONDITIONAL, DECLARATIVE	—	MANDATORY
10	JURISDICTION: Colorado; PARTY: M. Bentley, M. Schuett	MANDATORY, PERMISSIVE, CONDITIONAL, PROHIBITIVE	Governing Law	MANDATORY
11	—	MANDATORY, CONDITIONAL	—	MANDATORY
12	DATE: Dec. 8 2005, Dec. 7 2005	—	—	—
Document summary: 13 clauses · Parties: ENVISION, SIERRA · Dates: Dec. 8 & 7 2005 · Jurisdiction: Colorado · CUAD categories: 6				

Table 6: Stage 2 constraint profiles for the CUAD Distributor Agreement (same document as Table 5). DUR = Duration entity; MONEY, DATE, PARTY, LEGAL_REF, JURISDICTION are extracted entity types. Full outputs available on the project repository.

7.2 Negative Results and Failure Modes

Several failure modes were identified in qualitative analysis:

- **Obligation Weakening:** Even with MANDATORY prefixes, fine-tuned BART occasionally converts "shall" to "may," weakening binding obligations ($\sim$4% of mandatory clauses). The NLI gate catches most but not all cases.
- **Cross-Reference Loss:** Clauses containing intra-document references sometimes lose the reference in simplification, as BART treats it as less important than the primary obligation.
- **Chunking Artifacts:** Long clauses (>300 words) processed in overlapping chunks occasionally introduce repetition or awkward transitions ($\sim$8% of long-clause simplifications).
- **Persona Collapse for Short Clauses:** For very short clauses (<15 words), Flan-T5 produces nearly identical outputs across all three personas ($\sim$12% of test clauses).
- **Entity Boundary Errors:** The regex-based DATE extractor occasionally captures sentence-initial years as dates (<3% of clauses).

7.3 Qualitative Analysis

Table 11 shows three representative end-to-end simplification examples demonstrating the pipeline's behaviour across different clause types.

8 Challenges, Limitations, and Ethics

8.1 Implementation Challenges

- **Compute Constraints:** Training Legal-BERT+CRF on CUAD required 2$\times$T4 GPUs on Kaggle with gradient checkpointing and FP16 mixed precision.
- **Synthetic Data Quality:** LLaMA 3.3-70B occasionally generates simplifications that introduce new information. A quality filtering step detects and removes pairs where the reference is longer than the complex input or contains novel proper nouns.

Source	Text Content
Original (Clause 3)	THIS EXCLUSIVE DISTRIBUTOR AGREEMENT shall be effective as of Dec. 8, 2005, by and between LifeUSA/ Envision Health, Inc. ("ENVISION"), and Sierra Mountain Minerals, Inc. ("SIERRA"). ... Recitals: A. SIERRA is the manufacture and producer of a joint health product called "SierraSil" ... [cite: 166]
Stage 3 Simplified	THIS EXCLUSIVE DISTRIBUTOR AGREEMENT must be effective as of Dec. 8, 2005, between LifeUSA/ Envision Health, Inc. ("ENVISION"), and Sierra Mountain Minerals, Inc. ("SIERRA"). This agreement is based on the following facts: Recitals: A. SIERRA makes and sells a joint health product called "SierraSil" for human use. [cite: 166]

Table 8: Verbatim Stage 3 qualitative output for Clause 3 of the Distributor Agreement.

Metric	Layperson	Student	Professional
Target FKGL	7.0	11.0	15.0+
Achieved FKGL	7.9	11.8	14.3
FKGL Gap	0.9	0.8	0.7+
Post-adaptation Entailment	91%	93%	96%
Entity Pass-through Rate	98%	99%	100%

Table 10: Stage 5 persona adaptation results. FKGL gap $= |\text{target} - \text{achieved}|$; lower is better.

- **API Rate Limits:** Both Cerebras (Stage 3) and Groq (Stage 5) impose rate limits; the resumable checkpointing mechanism was essential.
- **NLI Label Ordering:** The `cross-encoder/nli-deberta-v3-large` model has a non-standard label ordering; the entailment index is detected dynamically from model config at runtime.

8.2 Limitations

8.3 Limitations

- **English Only:** The system is trained exclusively on English legal contracts.
- **Jurisdiction Bias:** The underlying dataset (CUAD) consists primarily of American commercial contracts. The system's generalisability to non-US legal frameworks or civil law jurisdictions remains untested.
- **Contract-Centric:** Optimised for commercial contracts (CUAD); legislative texts and court opinions may require separate adaptation.
- **No Human Evaluation:** Due to time constraints, the planned 50-clause human evaluation with clarity and fidelity annotation was not completed.
- **Threshold Sensitivity:** The NLI acceptance threshold $\tau = 0.85$ was selected based on a preliminary sweep and not extensively cross-validated.

8.4 Ethics and Responsible Deployment

Deploying an automated legal text simplification system raises several important ethical considerations:

- **Legal Advice Disclaimer:** Simplified legal text must never be presented as legal advice. Simplified text should always be presented alongside the original with a clear disclaimer.
- **Error Propagation:** The 6% human review queue after Stage 4 means approximately 1 in 17 clauses contains potentially unfaithful simplification, which can be consequential for high-stakes documents.
- **Bias in Synthetic Data:** LLM-generated simplifications may encode biases from the LLM's training data, potentially favouring one party type.
- **Accessibility vs. Over-Simplification:** The layperson persona deliberately sacrifices some precision for readability; users should be made aware of this trade-off.

9 Conclusion and Future Work

9.1 Conclusion

Juris-Clarify addresses a critical gap in legal NLP: the absence of formal semantic verification in legal text simplification. By combining a hierarchical clause segmenter (LegalBERT + CRF), a structured constraint extractor, a constraint-aware BART rewriter trained on LLM-generated synthetic data, an NLI-based verification loop, and a persona adapter (Flan-T5), the system provides both more readable outputs and meaningful formal guarantees of semantic faithfulness.

All five stages have been fully implemented and evaluated. Stage 1 achieves boundary F1 of 0.71 (WindowDiff 0.18), Stage 3 reduces FKGL by 6.3 grade levels with ROUGE-L of 0.58 and 89% entity preservation, Stage 4 increases NLI acceptance from 78% to 94%, and Stage 5 achieves

Clause Type	Original (excerpt)	Stage 3 Simplified	Layperson (Stage 5)
MANDATORY	"Acme Corp shall indemnify and hold harmless the Distributor from any claims arising from product defects…"	"Acme Corp must protect the Distributor from lawsuits about product problems…"	"Acme must pay for any lawsuits if their product causes problems."
CONDITIONAL	"…unless Distributor provides written notice of termination at least sixty (60) days prior to the expiration…"	"…unless the Distributor gives 60 days' written notice before the agreement ends…"	"…unless you send a written notice 60 days before it ends."
PROHIBITIVE	"…shall not directly or indirectly solicit, recruit or hire any employee of the Company for a period of eighteen (18) months following termination…"	"…may not hire Company employees for 18 months after the agreement ends…"	"…cannot try to hire the company's workers for 1.5 years after it ends."

Table 11: Qualitative end-to-end simplification examples across three clause types.

per-persona FKGL gaps within 0.9 grade levels. The ablation study confirms that each architectural component contributes measurably to the overall pipeline quality. The modular architecture allows each component to be independently evaluated, ablated, and improved.

9.2 Future Work

- **Human Evaluation:** Complete the 50-clause human evaluation with two annotators measuring clarity and fidelity, computing Cohen's κ for inter-annotator agreement.
- **Cross-Domain Extension:** Extend to legislative texts and court opinions using LEDGAR (750K provisions) and case law corpora.
- **Multilingual Adaptation:** Adapt the pipeline to French, German, and Spanish legal documents using multilingual LegalBERT variants.
- **Contrastive NLI Training:** Fine-tune the DeBERTa verifier on ContractNLI with contrastive examples for improved calibration.
- **Interactive Demo:** Deploy a web interface with clause-by-clause simplified outputs, persona selection, and entity highlighting.
- **Obligation-Type Aware Generation:** Extend Stage 3 to use separate BART decoder heads per obligation type.

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